Abstract

Pacific salmon populations are destabilizing across the North Pacific, with widespread declines in abundance, diversity, and body size threatening both ecosystems and human communities. Understanding their marine distribution is essential for effective management, yet current knowledge relies heavily on fisheries-dependent sampling. Pop-up satellite archival tags (PSATs) offer a fisheries-independent alternative, but their positive-only, autocorrelated data are unsuitable for traditional distribution modeling techniques. We propose adapting particle tracking methods to PSAT movement data through a novel approach we term species redistribution modeling (SRM). Rather than modeling where fish occur, SRMs model how fish move, then use those movement probabilities to identify basins of attraction - areas where fish are predicted to accumulate. Using PSAT data from Chinook salmon (Oncorhynchus tshawytscha) in the Gulf of Alaska, we will build a movement model and use it to identify these basins. To validate whether these basins correspond to areas of higher density, we will test their ability to predict Chinook salmon bycatch risk in the Gulf of Alaska groundfish fishery. Success will demonstrate that SRMs can serve as a fisheries-independent tool for modeling Pacific salmon marine distribution, with direct applications to bycatch risk assessment and marine spatial planning.

Introduction

Pacific salmon (Oncorhynchus spp) fuel food webs that sustain both wildlife and human communities throughout the North Pacific. Born in freshwater, these fish amass 99% of their body weight in the marine environment (Gende, 2004) and when they return to spawn and die, that marine bounty becomes fertilizer for their natal watersheds (Gende, 2004; Quinn, 2005). Those that don't return feed killer whales (Orcinus orca), Steller sea lions (Eumetopias jubatus), salmon sharks (Lamna ditropis), and other marine predators (Adams, 2016; Seitz, 2019). For human communities, Pacific salmon have provided nourishment and shaped cultures for millennia (Atlas, 2021). In Alaska alone, Pacific salmon constitute one third of the 34 million pound annual subsistence harvest (Fall, 2016) and hundreds of millions of pounds in commercial landings (NOAA, 2024).

Unfortunately, these populations are now destabilizing across the North Pacific, threatening the ecosystems and economies they sustain. Of the 53 populations of Pacific salmon in the continental United States, more than half are listed under the Endangered Species Act, and despite billions of dollars in recovery programs, none have been delisted (Ford, 2025). Across North America, return rates of Chinook salmon (Oncorhynchus tshawytscha) have declined threefold since 1978 (Welch, 2021), while Chum salmon (Oncorhynchus keta) returns in British Columbia's Central Coast region fell by 91% between 1960 and 2020 (Atlas, 2022). The Yukon River watershed saw Chinook salmon harvests drop 70% between 1998-2010 compared to the prior three decades (Krueger, 2013). Sockeye salmon (Oncorhynchus nerka) in the Fraser River hit such a low in 2009 that the Prime Minister of Canada established a committee to investigate. That return numbered 900,000 fish, yet by 2019 the return fell to 490,000 and by 2020 to just 290,000 (Beamish, 2022).

The picture grows more concerning when viewed through the lens of diversity. Between 1970 and the 2010s, Pacific salmon harvests actually increased 4.9-fold in Russia and 2.6-fold in the United States (Beamish, 2022). Combined with the aforementioned declines, this indicates an overall drop in diversity. In Puget Sound, Pink (Oncorhynchus gorbuscha) and Chum salmon made up approximately half of the stock in the 1970s; between 2005 and 2016 they made up 80% (Losee, 2019). The Skeena River tells a similar story through a different measure: the stabilizing effect of multiple populations - the portfolio effect - has weakened dramatically, with returns that were twice as stable as a single population between 1913-1923 dropping to only 1.1 times as stable by 2010-2017 (Price, 2020). Hatcheries, an early response to these declines, have not reversed the trend (McMillan, 2023). Of 102 papers reviewed by McMillan et al. on genetic diversity in populations with hatchery fish, 66 indicated declines.

Pacific salmon are also getting smaller. Chinook salmon in Alaska were 8% smaller in samples after 2010 than in samples before 1990, with Coho (Oncorhynchus kisutch), Chum, and Sockeye showing reductions of 3.3%, 2.4%, and 2.1% respectively (Oke, 2020). Part of this decline in size reflects a decrease in age at maturity. Before 1975, Alaskan Chinook salmon that had spent five years at sea represented 3-5% of returning spawners (Ohlberger, 2018), but by 2009 that proportion had dropped below 0.5%. This shift is especially concerning because larger, older fish are disproportionately important to stock productivity (Hixon, 2013).

Unraveling these population dynamics depends on understanding Pacific salmon's marine ecology as all species of Pacific salmon make extensive use of the marine environment to power their growth (Beamish, 2018). Chum salmon migrate to coastal waters immediately after emerging from their gravel beds, rear as juveniles, then range through the open ocean for up to six years before returning to spawn. Sockeye salmon spend more time in freshwater (1-3 years) but then follow a similar trajectory, spending up to three years at sea. Chinook salmon from rivers in Oregon regularly journey to the Gulf of Alaska during their up to four years in the ocean (Weitkamp, 2010). Pink salmon are perhaps the most impressive in their movements, migrating directly to the marine environment upon hatching and then traveling thousands of kilometers in their 18 months at sea. And while Coho salmon spend less time in the ocean than their relatives, they still manage to amass 99% of their weight in marine waters (Beamish, 2018). Clearly, understanding the marine ecology of these species is essential to managing them effectively.

However, our current understanding of their marine distribution has largely depended on sampling through the capture of fish at sea - leaving us with a partial picture. Studies of distribution in the North Pacific have come through high seas tagging studies conducted first by the International North Pacific Fisheries Commission and continued by the North Pacific Anadromous Fish Commission (Langan, 2024; Myers, 2007). While these studies have been invaluable in prying open the "black box" of Pacific salmon distribution, the spatiotemporal patchiness of sampling leaves the picture incomplete. Since the 1990s, for example, there have been fewer and fewer samples in the eastern North Pacific (Langan, 2024). Closer to the North American coast, Weitkamp et al. used recaptures of tagged fish to map the overall distribution of Coho and Chinook salmon by source region and age class (2002 & 2010), but because sampling relied on a patchwork of different fisheries, the analysis could only operate at coarse geospatial scales. Studies attempting finer resolution have been limited to modeling where fisheries and salmon intersect (DeFilippo, 2025; Shirk, 2023; Freshwater, 2021) - a particular problem in areas where Pacific salmon is bycatch and fishers are therefore incentivized to avoid it (DeFilippo, 2025).

Pop-up satellite archival tags (PSATs) provide one means to remove this dependency on fisheries data. Once attached to a fish, PSATs record the data required for environmental geolocation for a set period before releasing from the fish, surfacing, and transmitting their data through satellite communications (Wildlife Computer, 2025). Because data collection does not require fish recovery, PSAT data is fisheries independent. It is also a tool that has already been successfully applied to Pacific salmon (Seitz, 2024).

The data from PSATs, however, have two properties that make them unsuitable for traditional species distribution modeling techniques. First, PSATs provide positive-only samples. That is, a lack of records in a specific time and place does not indicate an absence of fish. Second, because the data captures individual fish movements, it is highly autocorrelated. The former property alone is not necessarily prohibitive; methods such as MaxEnt and EcoCast exist for working with positive-only samples (Phillips, 2005; Hazen, 2018). Both approaches, however, implicitly require that positive samples are representative of the underlying distribution - an assumption likely contradicted by both the autocorrelated nature of the data and the extraordinary range of Pacific salmon (Beamish, 2018). EcoCast also requires artificial generation of negative samples, introducing data not derived from measurement.

An alternative that has not yet been explored, is using the movement data from PSATs to adapt particle tracking techniques. Lagrangian particle tracking has already been used to understand distributions of larval dispersal (Quigley, 2024), turtle hatchling distribution (Le Gouvello, 2020; Harrison, 2021), and the movement of plastic debris (Lebreton, 2012; Onink, 2021). The approach seeds simulations with particles, projects their movements forward, and identifies areas where particles accumulate - also known as basins of attraction. For larvae and plastic, movements are guided passively by oceanic currents, winds, and other abiotic phenomena. Pacific salmon, by contrast, present a unique challenge in that they are active swimmers. Therefore tracking their potential movements requires explicit movement modeling. PSAT data is ideal for this because it captures both positive examples (observed movements) and negative examples (possible but unobserved movements) explicitly, enabling the use of traditional modeling techniques. In short, PSAT data can be used to build movement models that can then be used with particle tracking techniques to identify basins of attraction. We will refer to this methodology as species redistribution modeling, or SRMs.

While these basins indicate predicted areas of accumulation, there are other factors affecting fish distribution. Migrations create immigration and emigration events that may not deposit fish precisely where they may later preferentially accumulate. Preferred areas may also be associated with higher mortality, suppressing expected density levels. Therefore, to validate whether the basins identified by SRMs are useful indicators of relatively higher concentration, an independent data source is required.

Here is where fisheries-dependent data can come into play. If an SRM's predictions are meaningful, we would expect that areas where fish are predicted to accumulate would generally experience higher catch levels than areas fish are predicted to move away from. Passing this test would provide evidence that an SRM has not only identified attractors in fish movement, but hotspots in distribution as well. Therefore, running this test requires a case study with both movement data from which to build the SRM and catch data to validate it.

Chinook salmon in the Gulf of Alaska provides such a case study. A substantial body of PSAT movement data already exists for Chinook salmon in this region (Seitz, 2024), and an active groundfish fishery generates Chinook salmon bycatch data suitable for validation.

Therefore, our goal is to use this case study to understand whether SRMs can serve as a fisheries-independent means of modeling Pacific salmon distributions. To accomplish this, we will first develop a movement model using Chinook salmon PSAT data in the Gulf of Alaska, then use this model to perform species redistribution modeling, and finally assess the ability of the resulting basins to differentiate Chinook salmon bycatch risk.

Methods

Validated Tag Movement Data

Pop-up satellite tag data from Chinook salmon in the Gulf of Alaska will serve as the foundation for this analysis. Existing tag deployment data from Seitz (2024) will provide the core dataset, with fish captured by hook and line across multiple locations: Dutch Harbor, AK (n=20), Chignik, AK (n=16), Craig, AK (n=8), Homer, AK (n=20), Kodiak, AK (n=13), Yakutat, AK (n=16), Sitka, AK (n=15), and Central Bering Sea (n=3). Only healthy individuals meeting minimum size criteria (62-100cm fork length) were selected for tagging, and pop-up satellite archival tags were attached following established protocols (Seitz, 2024). Tagged fish were released at their capture locations.

From these deployed tags, time-series depth, temperature, and position data was collected once the tags surfaced. Tags were programmed to release from the fish, surface, and transmit collected data over satellite, with data transmitted in time-series format at 15 minute intervals. These measurements enabled light-based geolocation (using sunrise and sunset events along with a GMT clock to estimate position) with temperature-based corrections that refined estimates by matching recorded temperatures to known sea surface temperature fields. The exact algorithm used is a proprietary algorithm from Wildlife Computers (Wildlife Computers, 2025) that produces likely paths for individual fish.

We will then transform these positions to a spatial grid. Using the likely path estimates, an estimate of location for each day will be derived as the central tendency of those paths. Position estimates will then be converted to Uber H3 hexagonal grid cells at resolution 4 (~26km edge length), a hierarchical spatial indexing system that provides a spatial grid across the globe. Aggregation to H3 resolution 4 accommodates geolocation uncertainty while maintaining meaningful spatial resolution, creating spatially-standardized movement tracks.

With gridded positions established, tag records will be filtered to retain only post-acclimation movement data. Fish behavior immediately following tagging may reflect handling stress rather than natural movement. All movement data from the first seven days post-deployment will be excluded from analysis to allow fish to return to normal behavioral patterns.

Post-mortality records must also be screened and removed from the movement dataset. Constant depth readings (no variation in pressure measurements) indicate mortality with the tag either on the seafloor or at the surface (Seitz, 2019). Temperature stabilizing at the internal body temperature of known salmon predators — differing from ambient water temperature — indicates predation mortality. Michael Courtney and Andrew Seitz will aid in identifing which predator body temperatures to use based on their previous Chinook salmon mortality research (Seitz, 2019). Each fish's track will be scanned to identify mortality transition points based on these indicators, and any data following a detected mortality will be excluded from the dataset.

These validated trajectories will be visualized alongside a summary of data retention across filtering steps. A map showing all validated movement trajectories will demonstrate geospatial coverage across the Gulf of Alaska. A summary table will report data retention: total tags deployed, tags recovered, data points before/after acclimation filter, and data points before/after mortality filter. Individual-level statistics will show the distribution of track durations and data point counts per fish, demonstrating spatial extent of data and quantifying impact of quality control steps.

Homing Migration Classification and Filtering

Salmon homing migrations (return journeys to natal streams) must be separated from routine at-sea movements to avoid biasing our analysis of distribution preferences (Beamish, 2018). A model that distinguishes, across the dataset, between migration movements and non-migration movements will therefore be necessary.

Construction of this classifier requires a training dataset. Michael Courtney and Andrew Seitz will assist in labeling unambiguous migration and non-migration periods, as no standard methodology exists for this classification. Clear positive and negative examples will establish the training foundation, while the classifier will handle ambiguous intermediate cases.

Example trajectories with classified migration and non-migration segments will be visualized to illustrate this labeling. Maps will show trajectories with migration segments (red) and non-migration segments (blue) overlaid. Examples will be selected to illustrate the range of movement patterns observed in the labeled data.

From these labeled examples, temporal movement features will be derived using bidirectional time windows that examine both forward and backward in time from the point of interest. Window sizes will be constrained to one week or less to help retain data from fish with shorter tagging periods. Features will be extracted from movement data only (no environmental features), as migration should be identifiable from the nature of the movement alone. Feature types will include momentum (sustained directional movement), behavioral consistency (erratic versus steady patterns), and temporal continuity (persistence across the window).

A supervised classifier will then be trained to predict migration periods from these movement features. The model will be constructed to predict the probability that a specific step is part of a migration given features from the bidirectional temporal window. A classification threshold (operating point) will be selected where false negatives (migration points incorrectly labeled as non-migration) are expected to be below 5% on the validation dataset. The true negative rate (non-migration points correctly labeled) at that threshold will then be evaluated - higher is better, as it means more non-migration samples are retained without accidentally including migration data. Class balancing will be applied during training to ensure equal representation of migration and non-migration periods. Candidate models will include random forests, gradient-boosted trees (XGBoost), and neural networks, with flexibility to explore other architectures if needed.

Classifier performance will be demonstrated through standard evaluation metrics. A confusion matrix (table showing correct vs. incorrect classifications) will display true/false positives and negatives on the validation set. ROC (Receiver Operating Characteristic) curves, which plot true positive rate against false positive rate across all possible thresholds, will be generated for both training and testing sets to assess generalization and identify overfitting.

The trained classifier will then be applied to isolate non-migratory movements from the full dataset. All unlabeled time points will be classified using the selected operating point, and points classified as migration will be removed from the dataset.

Results of this filtering will be summarized in tabular form. A summary table will report data filtered: total time points, points classified as migration, and points classified as non-migration. Individual-level statistics will show the distribution of migration periods per fish.

Feature-Enriched Training Data

With validated non-migratory movement data in hand, a comprehensive list of candidate environmental features that could be used in movement modeling will be developed and categorized. An exhaustive feature list will be compiled from published literature and expert panel input (Andrew Seitz, Michael Courtney, Curry Cunningham) to ensure consideration of features beyond those typically used. Features will be categorized as: stationary (unchanging features like bathymetry), temporal (features that vary with time like temperature), or stateful (features that accumulate or have memory like productivity). Such features will be pulled from ocean models and remote sensing platforms such as Copernicus Marine Service and Google Earth. All features will be standardized to Uber H3 resolution 4 and daily temporal resolution for joining with the movement data.

This initial feature set will then be clustered by information content using vector embedding analysis. Highly correlated features can cause overfitting, so dimensionality reduction techniques will be used to identify redundant information. Vector embeddings use neural networks to learn non-linear relationships between features, unlike PCA (Principal Component Analysis) which only captures linear relationships. Embeddings will reduce the exhaustive feature set to a smaller set, allowing for more efficient exploration of the feature space. However, embedded features lose their physical interpretability. To restore interpretability, original features that best predict the embedded features will be identified. These representative features will become the modeling candidates.

Representative features will be normalized and scaled for neural network compatibility. Neural networks learn by adjusting weights based on feature values; features with larger numeric ranges can dominate the learning process if not normalized. Features will be rescaled to [0,1] if non-negative (e.g., depth, distance) or [-1,1] if containing negative values (e.g., temperature anomalies), ensuring all features contribute equally during model training.

Movement data will then be resampled to ensure balance across space, time, and individual fish. The study area will be divided into four spatial regions: Aleutian Island Chain, Kodiak to Yakutat, Southeast Alaska (to British Columbia border), and British Columbia and south. These spatial regions correspond with typical fisheries management boundaries in the Gulf of Alaska. Each spatial region will be further divided into seasonal blocks (Winter, Spring, Summer, Fall) as Chinook behavior varies seasonally (Gietzmann-Sanders, in review; Seitz, 2024). Finer temporal resolution would likely spread data too thinly. Samples will be collected into each space-time block with equal representation across all blocks (as possible). Within each block, data from each individual will be resampled with replacement to ensure all individuals in each area/time are equally represented.

From these balanced observations, examples will be created and joined to features to produce the final dataset. First, the 90th percentile range of movement per day per fish will be calculated. Then, training samples from each movement will include: origin location, actual destination, and all potential destinations (H3 resolution 4 cells) within the 90th percentile daily range to capture realistic movement options. Actual destinations will be labeled as "selected". These positions and times will then be joined to the corresponding features described above.

Movement Probability Model

The movement model will be posed as a classification problem, predicting where a fish will go based on the environmental features at potential destinations (including staying in place). Classification models output the probability of selecting each destination, which can be used to build a transition matrix (a table showing the probability of moving from any location to any other location). This transition matrix reveals how fish density at any point in the grid will redistribute in the next timestep - exactly what is needed for tracking population distribution. Note that this approach is more computationally efficient than traditional particle tracking, which simulates many individual fish trajectories. Instead of tracing thousands of individual movements, we will be able to trace overall density shifts through time.

A log-odds modeling framework will be used to build these movement models. With many potential destinations, each having its own environmental features, the total number of model inputs grows very large. High-dimensional feature spaces require exponentially more data to fit effectively - a problem known as the "curse of dimensionality" (Verleysen, 2005). Log-odds models address this by evaluating each destination's features independently rather than jointly, reducing dimensionality from (features × destinations) to just (features). This dramatically reduces data requirements and overfitting risk (Gietzmann-Sanders, in review). The mimic framework (Gietzmann-Sanders, in review), which implements log-odds modeling for choice prediction, will be employed.

As the first step, input data will be reorganized in preparation for training. Data will be split into training, testing, and validation sets. Each individual fish will be assigned to only one set (no fish split across sets), preventing data leakage where the model learns patterns from the same fish it will be tested on. Contrast resampling (Gietzmann-Sanders, in review), a technique required for log-odds modeling that structures data as pairwise comparisons, will be employed. If high biases towards specific regions/times in any of the datasets are observed after this step, unbiasing may be skipped earlier and applied at this point instead.

Models will then be trained and hyperparameter tuned across the feature set. Hyperparameter tuning (adjusting model settings like learning rate and network size) will be driven by performance on the validation set. The test set is held out until final evaluation to provide an unbiased estimate of model performance. Performance will be measured by negative log likelihood (how well the model's predicted probabilities match observed movements). The model that performs best on the validation set will be selected.

Feature predictivity will then be assessed by training models with incrementally added features. Starting with a null model (predicting movement without any features), features will be added one at a time. At each step, models with each remaining feature will be fit and the one that improves validation performance most will be selected. This will be continued until the full model is reached. This incremental approach quantifies how much predictive power each feature contributes beyond features already included. Each incremental model will be evaluated on the held-out test set for final performance assessment, and the most predictive model across the test set will be selected for subsequent analyses.

Performance metrics will be summarized across selected and incremental models. A performance table will compare predictive accuracy across the incremental models. Metrics will include training, validation, and testing set performance for the models to identify generalization (ability to predict on new data).

Relationships between environmental features, observed movements, and model predictions will be visualized. Plots will show how predicted movement probabilities vary with key environmental features to validate that model behavior matches ecological expectations. Comparison of predicted versus observed movement patterns across feature gradients will be included. These visualizations will allow better understanding of what the model is learning and how it relates to what is understood about the ecology and physiology of Chinook salmon.

Bycatch Risk Maps

Catch data from the Gulf of Alaska pollock (Gadus chalcogrammus) fleet will be processed and standardized for risk modeling. Bycatch will be binned to Uber H3 resolution 4 and daily temporal resolution to match the movement model's spatiotemporal scale. Two standardization approaches will be used: by effort alone (standard CPUE, or catch-per-unit-effort), and by effort weighted by depth-occupancy likelihood (how often fish are predicted to be at trawl depths) (Gietzmann-Sanders, in review). Depth weighting accounts for pollock trawls occurring primarily near the seafloor (De Robertis, 2006). From this, two datasets will be created: a binary catch/no-catch dataset to test whether any amount of bycatch can be predicted, and a normalized catch quantity dataset to test whether bycatch levels can be differentiated.

Bycatch data coverage and characteristics across space and time will be summarized. A summary table will show the number of fishing events, spatial coverage, and temporal coverage. Maps will show the distribution of bycatch events across the Gulf of Alaska, demonstrating data availability for training the risk model and identifying coverage gaps that may limit prediction reliability in certain regions or time periods.

Movement probability matrices will be generated and density tracking used to identify basins of attraction. Model features will be computed across the entire Gulf of Alaska for trawl data time periods ±1 month to enable rolling window analysis. These model features will be used to predict transition matrices per day. For each time point in the bycatch data, the analysis will move back 1 week and start with a uniform initial distribution (equal probability across all cells). This distribution will then be propagated forward using the transition matrices, revealing likely areas of fish accumulation. Specifically, the resulting density per H3 cell will be used as an "attractor index" for that cell (higher values indicating areas more likely to attract fish). This process will be repeated for 2 weeks prior, 3 weeks, and 4 weeks to assess performance across different time horizons (week to month), with flexibility to adjust windows based on findings. Each trawl event will ultimately be paired with attractor indices derived from these 1, 2, 3, and 4 week time windows.

Training data will then be constructed from paired fishing events. Given that attractor indices only have meaning relative to other nearby H3 cells, fishing events that are reasonably close in space and time (constrained by data availability) will be paired. Equal representation of three outcomes will be sought: both events have bycatch, only one has bycatch, and neither has bycatch. This balance prevents the model from defaulting to always predicting risk differences, ensuring it can identify when risk levels are actually similar. These pairs will then be divided into training, validation, and testing sets.

A decision tree will then be trained to predict relative bycatch risk using attractor-derived features. The problem will be set up as predicting which location in each pair is the better choice (results in less bycatch) given only the pair's attractor indices. Decision trees will be used for their simplicity and interpretability, enabling full explanation of the decision-making process; alternative models will be considered if predictive performance is insufficient. A loss function (used to guide model training) based on expected bycatch given the probabilities the model assigns to each decision will be used. Minimizing this loss creates a decision tree that minimizes expected bycatch over the training samples. The validation set will be used to explore features, model structures, and hyperparameters.

Bycatch risk model performance will then be evaluated on the held-out testing data. Performance metrics comparing predictions to observed bycatch on the testing set will be reported. Results will be reported relative to a 50/50 random baseline, representing the null hypothesis (what would be expected if the model had no predictive ability). This will demonstrate whether the model provides meaningful risk predictions beyond chance.

Assuming the model is predictive, interpretable spatial risk maps will be generated for specific time periods. The model will generate pairwise distances (difference in assigned probabilities) between all cells, which will be used to group similar areas. Agglomerative clustering (a method that progressively merges similar items) will group cells with similar risk levels using these distances, with flexibility to explore alternative clustering methods if needed. Clusters will be assigned colors to create discrete risk zones (minimally green/yellow/red for low/medium/high risk) that are immediately interpretable for decision-making. These visualizations will demonstrate the spatial structure of bycatch risk as predicted from movement patterns.

Success Criteria

Migration Classification Model

Goal: Ensure the migration classifier accurately distinguishes migration from non-migration periods while minimizing contamination of the non-migration training dataset.

Criteria: The classifier achieves a false negative rate below 5% while maintaining a true negative rate above 75% on the validation set.

Movement Probability Model

Goal: Develop a parsimonious and generalizable movement probability model that predicts Chinook salmon movement choices based on environmental features.

Criteria: The selected model demonstrates consistent performance across training, validation, and testing sets according to the negative log likelihood of the data given the predicted probabilities from the model.

Bycatch Risk Maps

Goal: Predict relative bycatch risk from movement-based fish accumulation patterns to support fishing effort decisions that minimize Chinook salmon bycatch.

Criteria: The decision tree model discriminates between high and low bycatch risk scenarios on held-out testing data at a level above a 50/50 random baseline.

Outcomes

Peer-Reviewed Publication

We will produce a paper detailing the results of this study that will be submitted for publication in an appropriate peer-reviewed journal.

Presentation

We will present the outcomes of this work at at least one professional meeting.

Open-Access Models and Tools

We will make all migration, movement, and decision models freely available and accessible online.

Open-Access Datasets

We will make our training, validation, and testing datasets available for those who would want to extend or replicate our findings.

References

- Adams J, Kaplan IC, Chasco B, Marshall KN, Acevedo-Gutiérrez A, Ward EJ. A century of Chinook salmon consumption by marine mammal predators in the Northeast Pacific Ocean. Ecological Informatics. 2016 July;34:44–51.
- Atlas WI, Ban NC, Moore JW, Tuohy AM, Greening S, Reid AJ, et al. Indigenous Systems of Management for Culturally and Ecologically Resilient Pacific Salmon (Oncorhynchus spp.) Fisheries. BioScience. 2021 Feb 15;71(2):186–204.
- Atlas WI, Wilson KL, Whitney CK, Moody JE, Service CN, Reid M, et al. Quantifying regional patterns of collapse in British Columbia Central Coast chum salmon (Oncorhynchus keta) populations since 1960. Can J Fish Aquat Sci. 2022 Dec 1;79(12):2072–86.
- Beamish RJ, editor. The ocean ecology of Pacific salmon and trout. Bethesda, MD: American Fisheries Society; 2018. 1147 p.
- Beamish R. The need to see a bigger picture to understand the ups and downs of Pacific salmon abundances. Browman H, editor. ICES Journal of Marine Science. 2022 May 23;79(4):1005–14.
- DeFilippo LB, Larson WA, Barry PD, Cunningham CJ, Langan JA, Garcia S, et al. Drivers and Dynamics of Salmon Bycatch in the Eastern Bering Sea Pollock Fishery. Fish and Fisheries. 2025 Nov;26(6):1107–21.
- Fall J, Godduhn A, Halas G, et al. Alaska Subsistence and Personal Use Salmon Fisheries 2016 Annual Report. 2016
- Ford MJ, Lindley ST, Barnas KA, Shelton AO, Spence BC, Weitkamp LA, et al. Abundance Trends of Pacific Salmon During a Quarter Century of ESA Protection. Fish and Fisheries. 2025 Nov;26(6):1087–106.
- Freshwater C, Anderson SC, Beacham TD, Luedke W, Wor C, King J. An integrated model of seasonal changes in stock composition and abundance with an application to Chinook salmon. PeerJ. 2021 Apr 20;9:e11163.
- Gende SM, Quinn TP, Willson MF, Heintz R, Scott TM. Magnitude and Fate of Salmon-Derived Nutrients and Energy in a Coastal Stream Ecosystem. Journal of Freshwater Ecology. 2004 Mar;19(1):149–60.
- Harrison CS, Luo JY, Putman NF, Li Q, Sheevam P, Krumhardt K, et al. Identifying global favourable habitat for early juvenile loggerhead sea turtles. J R Soc Interface. 2021 Feb;18(175):rsif.2020.0799, 20200799.
- Hazen EL, Scales KL, Maxwell SM, Briscoe DK, Welch H, Bograd SJ, et al. A dynamic ocean management tool to reduce bycatch and support sustainable fisheries. Sci Adv. 2018 May 4;4(5):eaar3001.
- Hixon MA, Johnson DW, Sogard SM. BOFFFFs: on the importance of conserving old-growth age structure in fishery populations. ICES Journal of Marine Science. 2014 Oct 1;71(8):2171–85.
- Krueger C, Bisson P, Bradford M, Clark B, Conitz J, Howard K, et al. AYK SSI Chinook Salmon Expert Panel.
- Quigley C, Roughan M, Chaput R, Jeffs A, Gardner J. Simulating larval dispersal across the distribution of the New Zealand green-lipped mussel: insights into connectivity and source-sink

dynamics. Mar Ecol Prog Ser. 2024 Mar 13;731:129–45.

- Quinn T. The behavior and ecology of Pacific salmon and trout. Second Edition. University of Washington Press, Seattle, Washington. 2018.
- Langan JA, Cunningham CJ, Watson JT, McKinnell S. Opening the black box: New insights into the role of temperature in the marine distributions of Pacific salmon. Fish and Fisheries. 2024 July;25(4):551–68.
- Lebreton LCM, Greer SD, Borrero JC. Numerical modelling of floating debris in the world's oceans. Marine Pollution Bulletin. 2012 Mar;64(3):653–61.
- Le Gouvello DZM, Hart-Davis MG, Backeberg BC, Nel R. Effects of swimming behaviour and oceanography on sea turtle hatchling dispersal at the intersection of two ocean current systems. Ecological Modelling. 2020 Sept;431:109130.
- Losee JP, Kendall NW, Dufault A. Changing salmon: An analysis of body mass, abundance, survival, and productivity trends across 45 years in Puget Sound. Fish and Fisheries. 2019 Sept;20(5):934–51.
- McMillan JR, Morrison B, Chambers N, Ruggerone G, Bernatchez L, Stanford J, et al. A global synthesis of peer-reviewed research on the effects of hatchery salmonids on wild salmonids. Fisheries Management Eco. 2023 Oct;30(5):446–63.
- Phillips SJ, Anderson RP, Schapire RE. Maximum entropy modeling of species geographic distributions. Ecological Modelling. 2006 Jan;190(3–4):231–59.
- Ohlberger J, Ward EJ, Schindler DE, Lewis B. Demographic changes in Chinook salmon across the Northeast Pacific Ocean. Fish and Fisheries. 2018 May;19(3):533–46.
- Oke KB, Cunningham CJ, Westley PAH, Baskett ML, Carlson SM, Clark J, et al. Recent declines in salmon body size impact ecosystems and fisheries. Nat Commun. 2020 Aug 19;11(1):4155.
- Onink V, Jongedijk CE, Hoffman MJ, Van Sebille E, Laufkötter C. Global simulations of marine plastic transport show plastic trapping in coastal zones. Environ Res Lett. 2021 June 1;16(6):064053.
- Price MHH, Moore JW, Connors BM, Wilson KL, Reynolds JD. Portfolio simplification arising from a century of change in salmon population diversity and artificial production. Journal of Applied Ecology. 2021 July;58(7):1477–86.
- Seitz AC, Courtney MB. Telemetry and Genetic Identity of Chinook Salmon in Alaska: Final Report. 2024
- Seitz AC, Courtney MB, Evans MD, Manishin K. Pop-up satellite archival tags reveal evidence of intense predation on large immature Chinook salmon (Oncorhynchus tshawytscha) in the North Pacific Ocean. Can J Fish Aquat Sci. 2019 Sept;76(9):1608–15.
- Shirk PL, Richerson K, Banks M, Tuttle V. Predicting bycatch of Chinook salmon in the Pacific hake fishery using spatiotemporal models. Zhou S, editor. ICES Journal of Marine Science. 2023 Jan 25;80(1):133–44.
- Welch DW, Porter AD, Rechisky EL. A synthesis of the coast-wide decline in survival of West Coast Chinook Salmon (Oncorhynchus tshawytscha, Salmonidae). Fish and Fisheries. 2021 Jan;22(1):194–211.
- Weitkamp LA. Marine Distributions of Chinook Salmon from the West Coast of North America Determined by Coded Wire Tag Recoveries. Trans Am Fish Soc. 2010 Jan;139(1):147–70.
- Weitkamp L, Neely K. Coho salmon (Oncorhynchus kisutch) ocean migration patterns: insight from marine coded-wire tag recoveries. Can J Fish Aquat Sci. 2002 July 1;59(7):1100–15.
- Wildlife Computers. Minipat. 2025
- Verleysen M, François D. The Curse of Dimensionality in Data Mining and Time Series Prediction. In: Cabestany J, Prieto A, Sandoval F, editors. Computational Intelligence and Bioinspired Systems.

Berlin, Heidelberg: Springer Berlin Heidelberg; 2005. p. 758–70. (Hutchison D, Kanade T, Kittler J, Kleinberg JM, Mattern F, Mitchell JC, et al., editors. Lecture Notes in Computer Science; vol. 3512).